

CoolChain

IoT-Enabled Cold Chain Last-Mile Logistics for Pharmaceuticals in Southeast Asia

CONFIDENTIAL INVESTMENT MEMORANDUM

Sector	Logistics
Stage	Seed
Geography	Asia
Funding Ask	USD 1,500,000

Executive Summary

CoolChain operates a temperature-controlled last-mile logistics network serving pharmaceutical distributors and hospital groups across Singapore, Malaysia, and Indonesia. Our IoT-enabled refrigerated vans and real-time monitoring dashboard ensure FDA-grade cold chain integrity from warehouse to delivery point. Founded in 2022, CoolChain has completed three paid pilots with major pharmaceutical clients and generated USD 120,000 in pilot revenue.

Problem

Southeast Asia loses an estimated USD 1.3 billion in pharmaceutical product annually due to cold chain failures during last-mile delivery. Existing logistics providers (DHL, FedEx) do not offer real-time temperature monitoring below vehicle level. Hospital and clinic deliveries require end-to-end cold chain documentation for regulatory compliance. The SE Asia pharmaceutical logistics market is valued at USD 2.8 billion and growing at 12% annually (Frost & Sullivan, 2023).

Solution

Refrigerated delivery vans equipped with CoolChain's proprietary IoT sensors log temperature, humidity, and GPS location every 30 seconds. A real-time monitoring dashboard alerts dispatchers to any cold chain deviation. Automated compliance certificates are generated per delivery for regulatory submission. Pricing: USD 18 per delivery plus a USD 500 monthly platform fee per client.

Founding Team

Name	Role	Background
David Tan	CEO & Co-Founder	8 years at DHL Temperature-Sensitive Division (Singapore); led cold chain operations
Rachel Lim	COO & Co-Founder	6 years operations management at Zuellig Pharma; specialist in pharmaceutical distrib

The founding team does not currently have a dedicated CTO. Technology development is led by David Tan with support from two contract engineers. CoolChain is actively recruiting a VP of Engineering to join by Q4 2024. Board currently comprises both co-founders only; we plan to add an independent board member post-funding.

Market Sizing

Total Addressable Market: USD 2.8 billion — SE Asia pharmaceutical logistics (Frost & Sullivan, 2023). Serviceable Addressable Market: USD 420 million — cold chain last-mile segment in Singapore, Malaysia, and Indonesia. Serviceable Obtainable Market: USD 21 million — hospital groups and large pharma distributors requiring real-time monitoring compliance documentation.

Revenue Model & Traction

Per-delivery fee model: USD 18 per cold chain delivery plus USD 500 monthly SaaS platform fee. Clients commit to a minimum of 200 deliveries per month under 12-month service agreements.

Metric	Current Value
Pilot Revenue (12 months)	USD 120,000
Active Pharmaceutical Clients	3 (pilot agreements)
Deliveries Completed	6,800
Average Revenue per Delivery	USD 17.65
Cold Chain Deviation Rate	0.3% (industry benchmark: 2.1%)
Net Promoter Score	72 (from client surveys, Q2 2024)

Competitive Differentiation

Competitor	Key Weakness vs CoolChain
DHL Medical Express	No real-time IoT monitoring; no automated compliance certificates
Zuellig Pharma Logistics	Full-distribution model; no last-mile-only API offering for hospitals
ColdEx (India)	India-focused; not licensed for SE Asia regulatory requirements

Milestones

Date	Milestone
Jan 2022	Company incorporated in Singapore; SGD 200K angel round
Jun 2022	First van deployed; Novartis Singapore pilot launched
Feb 2023	Two additional pharma clients onboarded (Zuellig, DKSH)
Oct 2023	IoT monitoring dashboard v2.0 launched; USD 120K revenue achieved
Q3 2024	Seed round close; fleet expansion to 15 vans; Malaysia market entry

ESG, Governance & Financial Projections

Environmental Commitments

CoolChain is committed to reducing its carbon footprint. By optimising delivery routes using our proprietary routing algorithm, we significantly reduce unnecessary mileage and fuel consumption compared to unoptimised logistics operations. We are evaluating a transition to electric refrigerated vehicles in our fleet by 2026. Our goal is to reduce carbon emissions across our operations as we scale, though we are still in the process of establishing a formal emissions measurement baseline.

Social Commitments

All CoolChain van drivers are direct employees (not gig workers) and receive above-minimum-wage compensation with healthcare benefits. CoolChain's operations ensure pharmaceutical access to underserved hospital groups in rural Indonesia. We currently employ 11 people across Singapore and Malaysia.

Governance

CoolChain's board currently comprises both co-founders. An independent director with pharmaceutical logistics expertise will be appointed post-funding. Financial accounts are prepared quarterly by an accredited Singaporean CA firm (Crowe Singapore).

Financial Projections

CoolChain projects strong revenue growth as the fleet expands into Malaysia and Indonesia following the seed round close. Year 3 revenue of USD 4.8 million is achievable given the large addressable market and limited direct competition in the IoT-enabled cold chain niche.

Metric	Year 1 (2024)	Year 2 (2025)	Year 3 (2026)
Fleet Size (vans)	15	35	80
Active Clients	8	18	40
Monthly Deliveries	3,200	7,500	18,000
Revenue (USD)	690,000	1,800,000	4,800,000
Gross Margin	48%	54%	61%

Use of USD 1.5M Seed Round: 55% fleet expansion (6 new refrigerated vans), 25% IoT hardware and software development, 12% Malaysia market entry costs, 8% working capital.